Decline Curve Fitting

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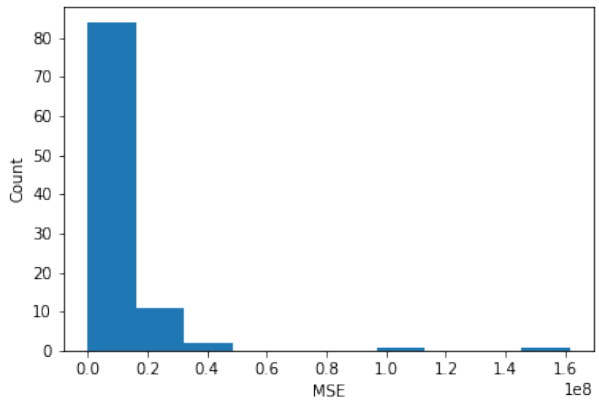
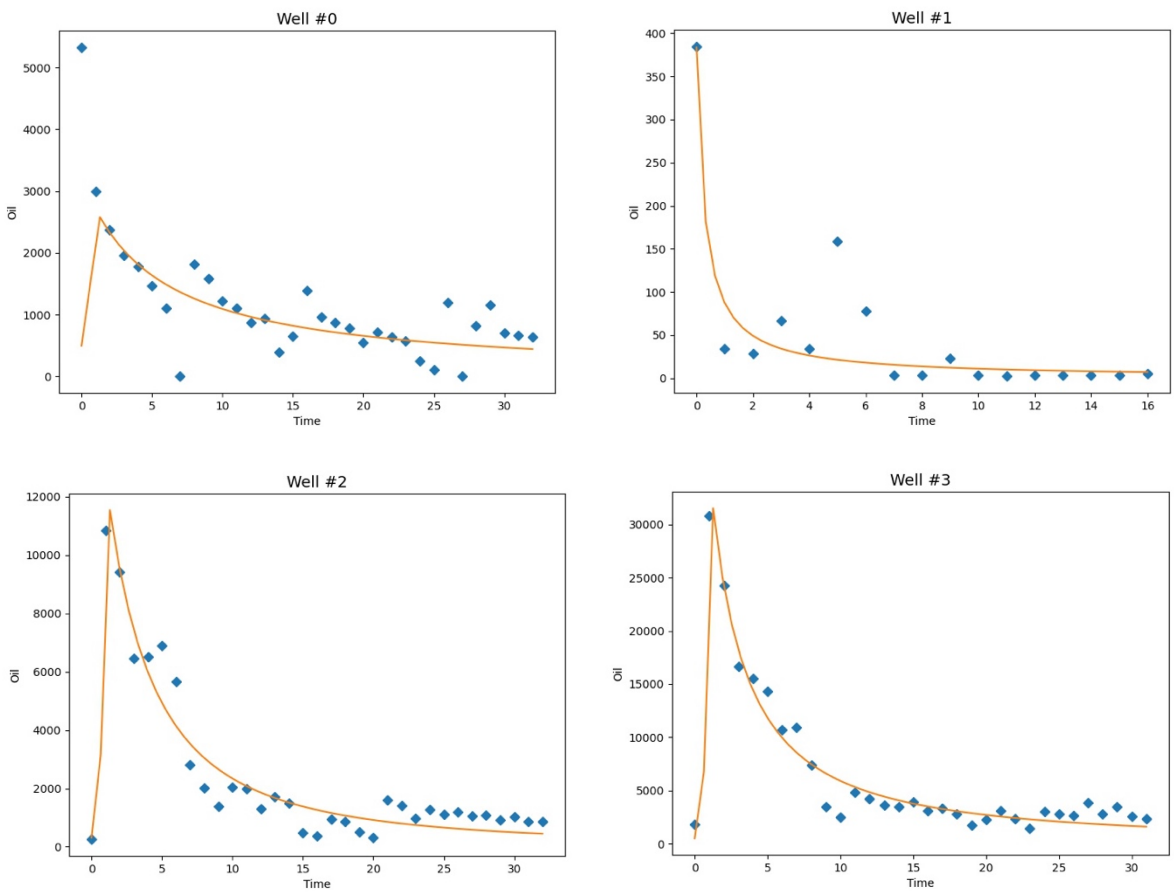
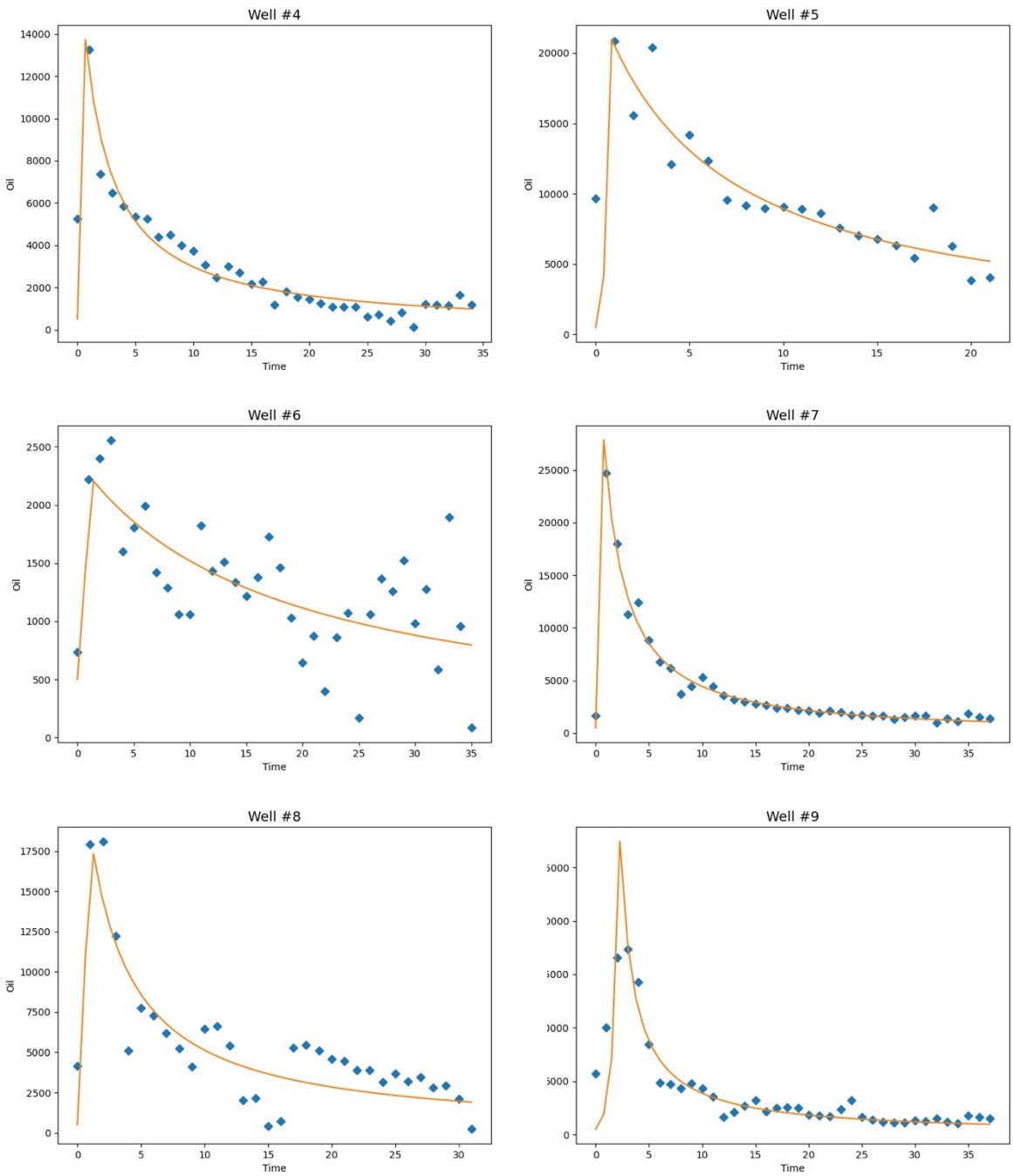
1. The time it takes to finish the 100 wells in total is 94s.
2. Below is the histogram plot of mean squared error (MSE) for all wells:



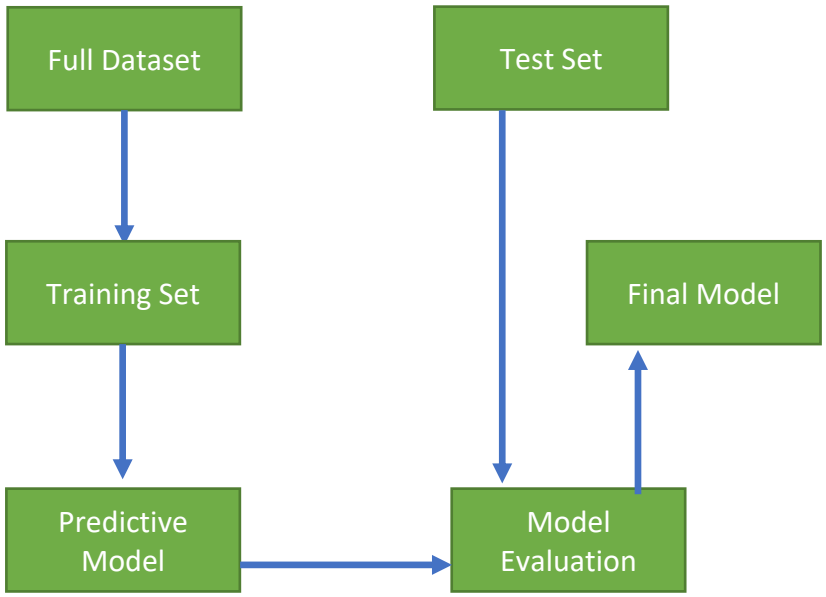
3. Below is the curve fit for the first 10 wells:



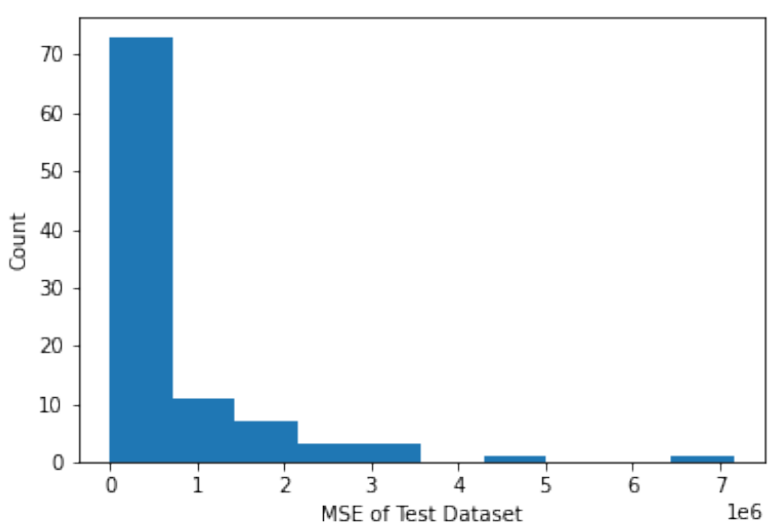


4. Loss function choice and why:
   I chose mean squared error (MSE) as loss function for three reasons:
   - The loss function value should be positive, so we can minimize loss
   - This loss function is more differentiable than the sum of absolute error(SAE)
   - MSE will decrease when dataset size increases and curve fits better, so it is better than sum of squared error(SSE)

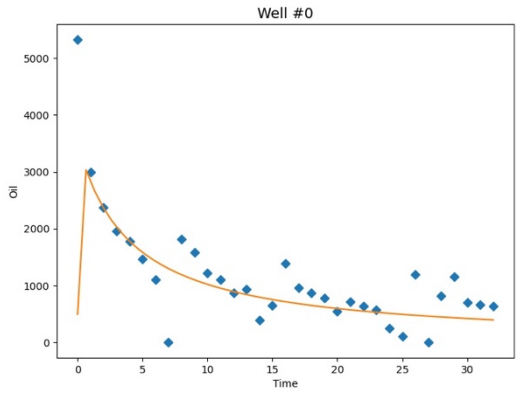
5. Outlier detection algorithm:
   For outlier detection, in this project I remove wells that have less than or equal to 5 data points and remove data points with 0 oil production.
6. Evaluation schema, and why and results
   The evaluation schema is defined as below.

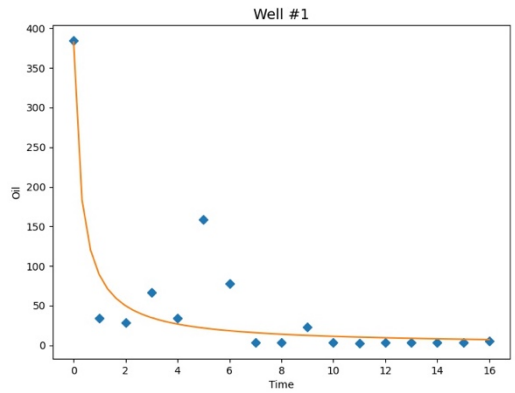


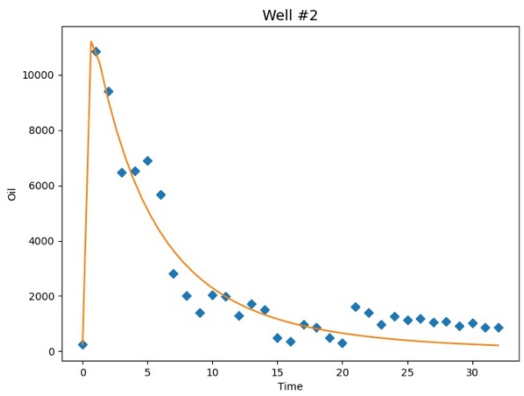
The full dataset is divided into 2 subsets, the training set and the test set. The training set is a subset of the dataset to build predictive models. I pick the first 80% of the dataset based on time. Test set is a subset of the dataset to assess the likely future performance of a model, which is the last 20% of the dataset based on time. The reason for this division is that we would like to predict the oil production based on the historical data. The MSE histogram show the algorithm behaves quite well.

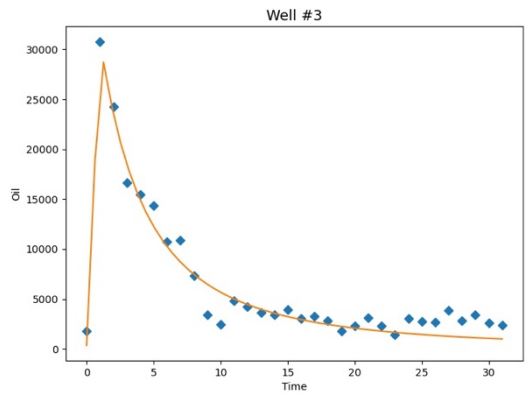


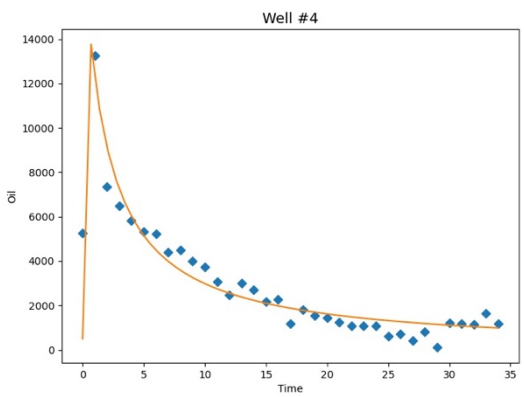
The following figures show the prediction power of the algorithm with 80% of the full dataset(training set). We can tell the differential evolution provides a good parameter estimate, and the fitting model behaves well.

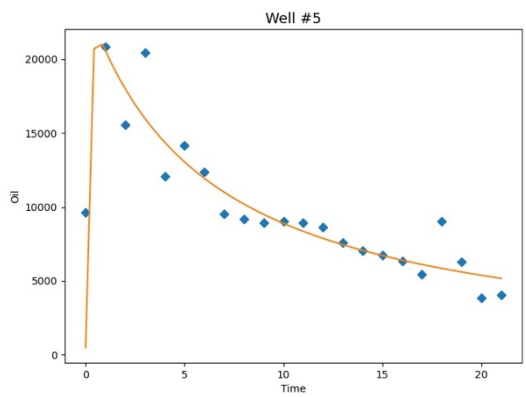


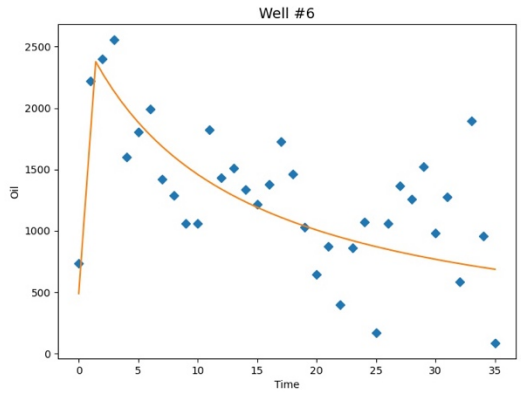


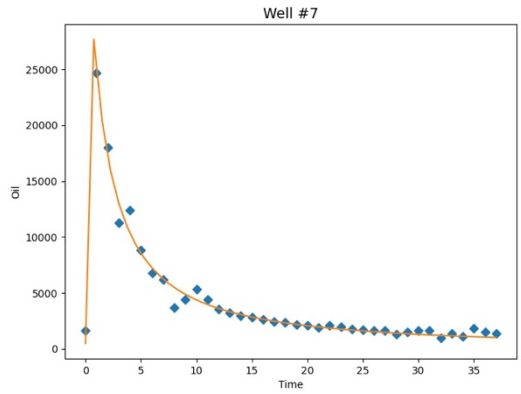


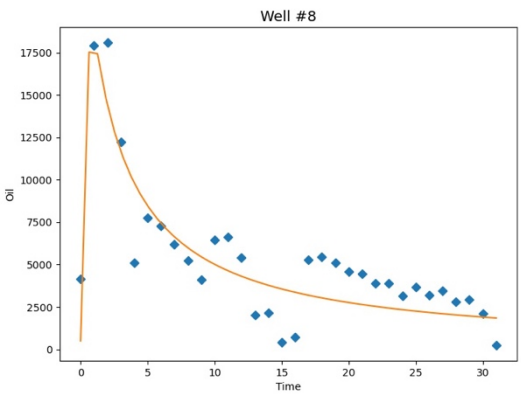


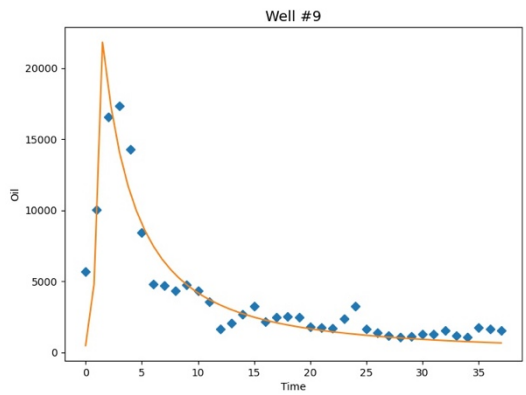





Well #6
Oil
Time


Well #7
Oil
Time


Well #8
Oil
Time


Well #9
Oil
Time